Exp No:13

Experiment to handle missing values and inappropriate data in a given data set

AIM:

Understand the procedure to handle missing and inappropriate data

PROGRAM:

```
import numpy as np
import pandas as pd
df=pd.read_csv("Hotel.csv")
df
df.duplicated()
df.info()
df.drop_duplicates(inplace=True)
df
len(df)
index=np.array(list(range(0,len(df))))
df.set_index(index,inplace=True)
index
df
df.drop(['Age_Group_2'],axis=1,inplace=True)
df
df.loc[df["CustomerID"] < 0, "CustomerID"] = np.nan
df.loc[df["Bill"] < 0, "Bill"] = np.nan
df.loc[df["EstimatedSalary"] < 0, "EstimatedSalary"] = np.nan
df
df.loc[(df["NoOfPax"] < 1) | (df["NoOfPax"] > 20), "NoOfPax"] = np.nan
df
df.Age_Group.unique()
df.Hotel.unique()
df["Hotel"] = df["Hotel"].replace("Ibys", "Ibis")
df["FoodPreference"].unique()
df.FoodPreference.replace(['Vegetarian','veg'],'Veg')
df.FoodPreference.replace(['non-Veg'],'Non-Veg')
df["EstimatedSalary"] =
df["EstimatedSalary"].fillna(round(df["EstimatedSalary"].mean()))
df["NoOfPax"] = df["NoOfPax"].fillna(round(df["NoOfPax"].median()))
df["Rating(1-5)"] =
df["Rating(1-5)"].fillna(round(df["Rating(1-5)"].median()))
```

```
df["Bill"] = df["Bill"].fillna(round(df["Bill"].mean()))
```

OUTPUT:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary	Age_Group_2
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	Lemon Tree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	Lemon Tree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	Lemon Tree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	Non-Veg	6755	4	87777	30-35

	0
0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	True
10	False

dtype: bool

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     object
 4   FoodPreference   11 non-null     object
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group_2      11 non-null     object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary	Age_Group_2
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	Lemon Tree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	Lemon Tree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	Lemon Tree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	Non-Veg	6755	4	87777	30-35

array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary	Age_Group_2
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25

1	2	30-35	5	Lemon Tree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	Lemon Tree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	Lemon Tree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	10	30-35	5	RedFox	Non-Veg	6755	4	87777	30-35

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000

3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	Ibys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	Non-Veg	6755	4	87777

	CustomerID	Age_Group	Rating (1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4	Ibis	veg	1300.0	2	40000.0
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3	59000.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2	30000.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2	120000.0
4	5.0	35+	3	Ibis	Vegetarian	989.0	2	45000.0
5	6.0	35+	3	Ibys	Non-Veg	1909.0	2	122220.0

6	7.0	35+	4	RedFox	Vegetarian	1000.0	-1	21122.0
7	8.0	20-25	7	LemonTree	Veg	2999.0	-10	345673.0
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3	NaN
9	10.0	30-35	5	RedFox	Non-Veg	6755.0	4	87777.0

	CustomerID	Age_Group	Rating (1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4	Ibis	veg	1300.0	2.0	40000.0
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3	Ibis	Vegetarian	989.0	2.0	45000.0
5	6.0	35+	3	Ibys	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4	RedFox	Vegetarian	1000.0	NaN	21122.0
7	8.0	20-25	7	LemonTree	Veg	2999.0	NaN	345673.0

8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0	NaN
9	10.0	30-35	5	RedFox	Non-Veg	6755.0	4.0	87777.0

array(['20-25', '30-35', '25-30', '35+'], dtype=object)
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
array(['veg', 'Non-Veg', 'Veg', 'Vegetarian'], dtype=object)

	FoodPreference
0	Veg
1	Non-Veg
2	Veg
3	Veg
4	Veg
5	Non-Veg
6	Veg
7	Veg
8	Non-Veg
9	Non-Veg

dtype: object

	CustomerID	Age_Group	Rating (1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4	Ibis	Veg	1300.0	2.0	40000.0

1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	7	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5	RedFox	Non-Veg	6755.0	4.0	87777.0

RESULT:

The experiment has been carried out successfully